DATABASE DESIGN

What is Database Design?

- Database design can be defined as a **collection of tasks or processes** that enhance the designing, development, implementation, and maintenance of enterprise data management system

- Designing a proper database **reduces the maintenance cost** thereby improving data consistency

- Cost-effective measures are greatly influenced in terms of **disk storage space**. Therefore, there has to be a brilliant concept of designing a database.

- The designer should follow the constraints and decide how the elements correlate and what kind of data must be stored.

Objectives of Database Design

- To produce physical and logical design models of the proposed database system

- The logical model is primarily concentrated on the requirements of data and the considerations must be made in terms of monolithic considerations and hence the stored physical data must be stored independent of the physical conditions

- The physical database design model includes a translation of the logical design model of the database by keeping control of physical media using hardware resources and software systems such as Database Management System (DBMS).

Importance of Database Design

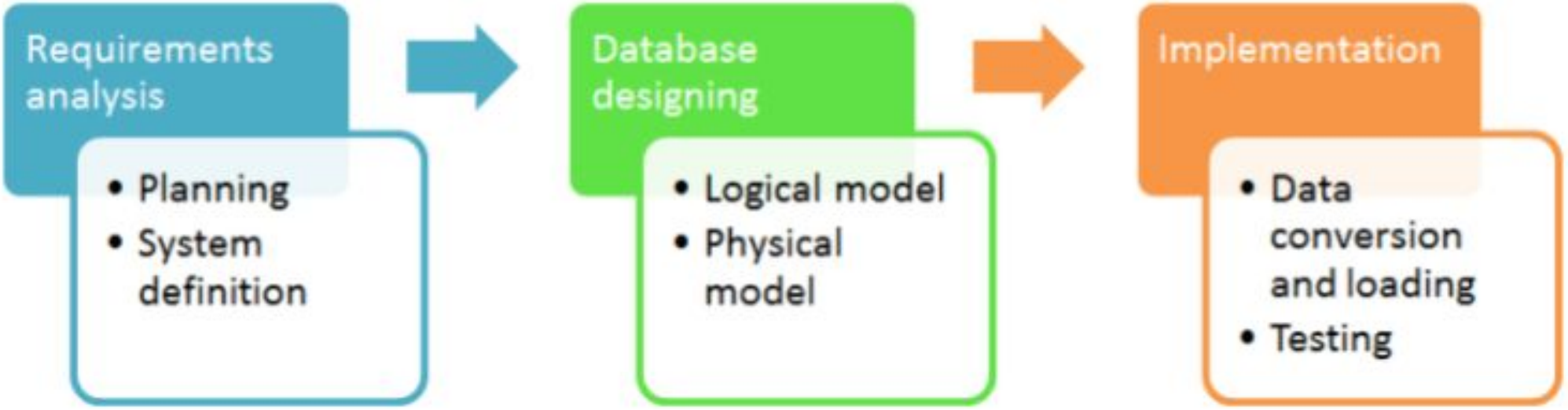
1. Database designs provide **the blueprints** of how the data is going to be stored in a system. A proper design of a database highly affects the overall performance of any application.

2. The designing principles defined for a database give a clear idea of the behavior of any application and how the requests are processed.

3. A proper database design meets all the requirements of users.

4. The processing time of an application is greatly reduced if the constraints of designing a highly efficient database are properly implemented.

Life Cycle of Database Design



i. Requirement Analysis

Planning has to be done on what are the basic requirements of the project under which the design of the database has to be taken forward.

They can be defined as:-

- **Planning -** This stage is concerned with planning the entire DDLC (**Database Development Life Cycle**). The strategic considerations are taken into account before proceeding.

- **System definition -** This stage covers the **boundaries and scopes** of the proper database after planning.

ii. Database Designing

The next step involves designing the database considering the user-based requirements and splitting them out into various models so that load or heavy dependencies on a single aspect are not imposed.

Therefore, there has been some **model-centric approach** and that's where logical and physical models play a crucial role.

- **Physical Model** - The physical model is concerned with the practices and implementations of the logical model.

- **Logical Model** - This stage is primarily concerned with developing a model based on the proposed requirements. The entire model is designed on paper without any implementation or adopting DBMS considerations.

Database Design Process

The process of designing a database carries various conceptual approaches that are needed to be kept in mind.

An ideal and well-structured database design must be able to:

1. Save disk space by eliminating redundant data.
2. Maintains data integrity and accuracy.
3. Provides data access in useful ways.
4. Comparing Logical and Physical data model

a. Logical Data Model

A logical data model **describes the data in as many details as possible, without having to be concerned about the physical implementations in the database**.

Features of logical data model might include:

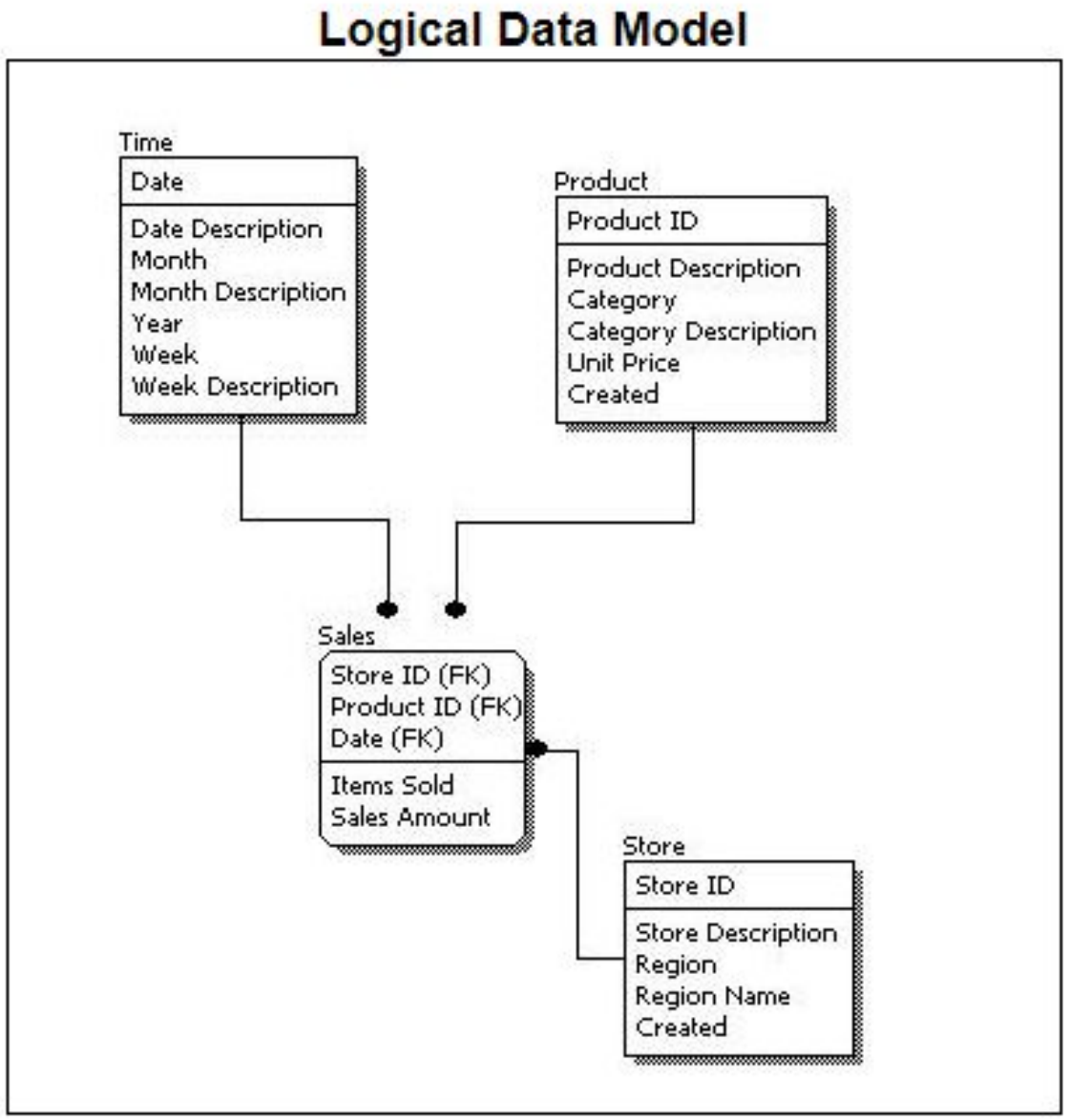
1. All the **entities and relationships** amongst them.
2. Each entity has well-specified **attribute**s.
3. The **primary key for each entity** is specified.
4. **Foreign keys** which are used to identify a relationship between different entities are specified.
5. **Normalization** occurs at this level.

A logical model can be designed using the following approach:

1. Specify all the entities with primary keys.
2. Specify concurrent relationships between different entities.
3. Figure out each entity attributes
4. Resolve many-to-many relationships.
5. Carry out the process of normalization.

After following the above approach, critically examine the design based on requirement gathering. If the above steps are strictly followed, there are chances of creating a highly efficient database design that follows the native approach.

a. Logical Data Model (contd...)



b. Physical Data Model

A Physical data model represents the approach or **concept of designing the database**.

The main purpose of the physical data model is to show all the **structures** of the table including the **column name, column data type, constraints, keys(primary and foreign)**, and **the relationship among tables**.

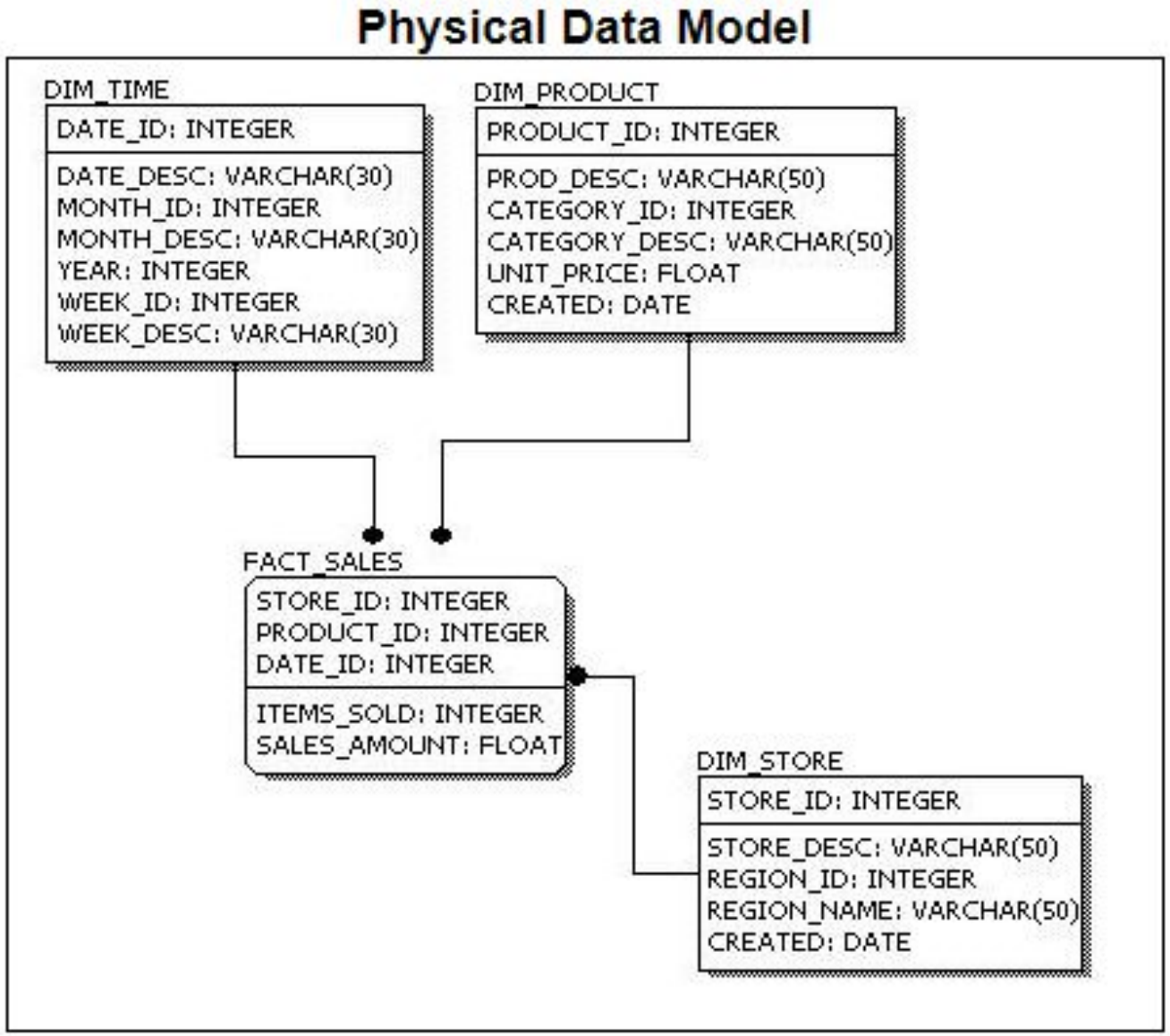
The following are the features of a physical data model:

1. Specifies all the columns and tables.
2. Specifies foreign keys that usually define the relationship between tables.
3. Based on user requirements, de-normalization might occur.
4. Since the physical consideration is taken into account so there will straightforward reasons for difference than a logical model.
5. Physical models might be different for different RDBMS. For example, the data type column may be different in MySQL and SQL Server.

While designing a physical data model, the following points should be taken into consideration:

6. Convert the entities into tables.
7. Convert the defined relationships into foreign keys.
8. Convert the data attributes into columns.
9. Modify the data model constraints based on physical requirements.

b. Physical Data Model (contd…)



iii. Implementation

The last step covers the implementation methods and checking out the behavior that matches our requirements.
It is ensured with continuous integration and testing of the database with different data sets and conversion of data into machine understandable language.
The manipulation of data is primarily focused on these steps where queries are made to run and check if the application is designed satisfactorily or not.

- **Data conversion and loading -** This section is used to **import and convert data from the old to the new system.**

- **Testing -** This stage is concerned with **error identification in the newly implemented system.** Testing is a crucial step because it checks the database directly and compares the requirement specifications.